THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** west to east	Counter-clockwise from above the North Pole.
2	**(a)** 21 June	Summer solstice.
3	**(b)** 27 days	Orbital period.
4	**(b)** Moon	Moon blocks Sun.
5	**(d)** 24 hours	It causes day and night.
6	**(b)** 22 December	Winter solstice.
7	**(d)** 29.5 days	Lunar month.
8	**(b)** new moon day	Aligned.
9	**(c)** 365 and a quarter days	One year.
10	**(d)** 21 March and 23 September	Equinoxes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rotation: spin on its axis (1); Revolution: motion around the Sun (1).	1+1
12	21 June (1); 22 December (1).	1+1
13	At new moon (1); Moon between Sun and Earth (1).	1+1
14	Earth rotates (1); Half faces the Sun (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Earth rotates (1); Side facing Sun has day (1); Other side has night (1).	1+1+1
16	Earth between Sun and Moon (1); Earth's shadow on the Moon (1); At full moon (1).	1+1+1
17	Orbit of Moon is tilted (1); Alignment is needed (1); Occurs rarely (1).	1+1+1
	OR: Axis tilted (1); Earth revolves around Sun (1); Different amounts of sunlight (1).	1+1+1
18	Year is 365.25 days (1); Quarter days add up (1); Extra day every 4 years (1).	1+1+1
	OR: Solar: Moon blocks Sun (1); Lunar: Earth's shadow on Moon (1); Solar at new moon, lunar at full moon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rotation: day and night (1); 24 hours (1); Revolution: year (1); 365.25 days (1); Seasons with tilt (1).	1+1+1+1+1
	OR: No atmosphere (1); Craters (1); Weaker gravity (1); Extreme temperatures (1); No water seen on surface (1).	1+1+1+1+1
20	Axis tilted 23.5° (1); June: northern hemisphere tilted to Sun (1); Longer days, direct rays (1); December: southern tilted (1); Reverse seasons (1).	1+1+1+1+1
	OR: The Sun rises in the east (1); The Earth rotates from west to east (1); The Sun appears to move from east to west (1); One rotation takes 24 hours (1); Day and night repeat (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** new moon	Amavasya.
2	**(b)** Earth	Earth's shadow falls on the Moon.
3	**(b)** rotation of the Earth	Half of the Earth faces the Sun.
4	**(c)** 23.5° N	Northern boundary of the tropics.
5	**(a)** full moon	Purnima.
6	**(c)** full moon day	Aligned.
7	**(b)** tilt of the Earth's axis and its revolution	Different parts get different sunlight.
8	**(b)** 23.5° S	Southern.
9	**(a)** increases (waxing)	Waxing.
10	**(d)** it can damage the eyes	Harmful rays.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Day and night equal (1); 21 March and 23 September (1).	1+1
12	At full moon (1); Earth between Sun and Moon (1).	1+1
13	Tilt of the Earth's axis (1); Revolution around the Sun (1).	1+1
14	Different shapes of bright part (1); Repeat in about a month (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hemisphere tilted towards the Sun (1); Sun above horizon for longer (1); Opposite in winter (1).	1+1+1
16	Rays more direct (1); Longer days (1); More heat absorbed (1).	1+1+1
17	Tilt (1); One tilted towards the Sun (1); The other away (1).	1+1+1
	OR: New moon: not seen (1); Waxing to full moon (1); Waning back to new moon (1).	1+1+1
18	Sun does not set for months (1); Sun does not rise for months (1); Near poles (1).	1+1+1
	OR: We see its lit part from different positions (1); Moon orbits Earth (1); Phases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	New moon: between Sun and Earth (1); First quarter (1); Full moon: Earth between (1); Last quarter (1); Cycle 29.5 days (1).	1+1+1+1+1
	OR: Sun: light and heat (1); Day-night cycles (1); Seasons for crops (1); Moon: lunar calendar (1); Festivals by phases (1).	1+1+1+1+1
20	Solar eclipse: Moon's shadow on Earth (1); Lunar: Earth's shadow on Moon (1); Umbra and penumbra (1); Do not view the Sun directly (1); Use safe filters (1).	1+1+1+1+1
	OR: Equator: about 12 h (1); Higher latitudes vary (1); Poles: months of day/night (1); Tilt effect (1); Summer longer (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 23.5°	From the vertical to the orbital plane.
2	**(d)** winter	Opposite.
3	**(a)** decreases (waning)	Waning.
4	**(c)** a pinhole projection or special filters	Indirect method.
5	**(b)** equator	0° latitude.
6	**(b)** direct (vertical)	Hotter.
7	**(b)** reflects sunlight	Non-luminous.
8	**(c)** umbra	Total eclipse region.
9	**(c)** leap year	366 days.
10	**(d)** slanting	Spread over more area.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Harms eyes (1); Use safe methods (1).	1+1
12	Has 366 days (1); February has 29 days (1).	1+1
13	Rotation equals revolution time (1); So same face points to Earth (1).	1+1
14	About 27 days (1); Around Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Moon reflects sunlight (1); Half always lit (1); We see different portions (1).	1+1+1
16	Earth rotates west to east (1); Sun appears to move opposite (1); Relative motion (1).	1+1+1
17	About 12 hours all year (1); Little seasonal change (1); Sun overhead (1).	1+1+1
	OR: Moon between Sun and Earth (1); Shadow falls on Earth (1); Sun blocked (1).	1+1+1
18	Make a small hole in card (1); Project Sun's image on paper (1); Do not look at the Sun (1).	1+1+1
	OR: Lies close to Earth's axis (1); Earth spins about it (1); Seems to stay (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	21 March, 23 September: equal (1); 21 June: longest day (north) (1); 22 December: shortest (1); Sun's overhead position changes (1); Seasons (1).	1+1+1+1+1
	OR: Torch as Sun (1); Rotate globe (1); Lit side is day (1); Dark side is night (1); Mark a place and follow (1).	1+1+1+1+1
20	Earth is closest in January (1); Yet north is cold (1); Tilt is the cause (1); Opposite seasons in two hemispheres (1); Distance varies little (1).	1+1+1+1+1
	OR: Lamp as Sun (1); Ball as Moon (1); Move around the observer (1); Lit part changes (1); Draw phases (1).	1+1+1+1+1

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- Writing that the Moon gives its own light; it only **reflects** sunlight.
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